Yihan Gao

Department of Aeronautical and Aviation Engineering
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EDUCATION

The Hong Kong Polytechnic University — Hong Kong, China
Ph.D. in Transportation — Sep 2023 - Aug 2027 (expected)
Department of Aeronautical and Aviation Engineering; Supervisor: Prof. Wei Liu

Central South University — Changsha, China
Master's Degree — Sep 2019 - Jun 2022
School of Traffic & Transportation Engineering

Central South University — Changsha, China
Bachelor's Degree — Sep 2015 - Jun 2019
School of Traffic & Transportation Engineering

RESEARCH INTERESTS

Methodologies
1. Transport network modeling and optimization
2. Equilibrium analysis and pricing
3. Robust optimization under uncertainty

Application Areas
1. Electric bus fleets and charging infrastructure
2. Shared parking, charging and vehicle-to-grid services
3. Railway seat allocation and service planning

RESEARCH EXPERIENCE

The Hong Kong Polytechnic University — Dec 2022 - Aug 2023
Research Assistant
Department of Aeronautical and Aviation Engineering

The Hong Kong Polytechnic University — Jan 2022 - Apr 2022
Visiting Student
Department of Aeronautical and Aviation Engineering

WORKING PAPERS

[1] **Gao, Y.**, and Liu, W. (2026). On the shared bus charging system with public access. *Submitted to Transportation Science.*

JOURNAL PUBLICATIONS

[1] **Gao, Y.**, and Liu, W. (2026). Robust planning for bus fleet electrification and charging facility deployment. *Transportation Research Part E: Logistics and Transportation Review*, 213, 104970.

[2] Zhang, Z., **Gao, Y.**, Zhang, F., and Liu, W. (2026). Parking management and operation: A bibliometric analysis and comprehensive review. *Transport Economics and Management*, 4, 14-37.

[3] Xu, J., Deng, L., **Gao, Y.**, and Liu, W. (2026). Joint optimization of pricing, seat allocation and overbooking for high-speed railway system under demand uncertainty. *Transportation Research Part C: Emerging Technologies*, 183, 105492.

[4] **Gao, Y.**, Liu, W., Zhang, Z., and Du, B. (2025). Managing parking and charging services for electric autonomous vehicles considering vehicle-to-grid facilities. *Transportation Research Part A: Policy and Practice*, 201, 104680.

[5] Xu, G., Zhou, P., Liu, W., and **Gao, Y.** (2025). Seat allocation optimization for railway systems with equity considerations. *Transportmetrica B: Transport Dynamics*, Article 2448707.

[6] Xu, G., Liu, Y., **Gao, Y.**, and Liu, W. (2023). Integrated optimization of train stopping plan and seat allocation scheme for railway systems under equilibrium travel choice and elastic demand. *Transportation Research Part E: Logistics and Transportation Review*, 177, 103231.

[7] Xu, G., **Gao, Y.**, and Liu, W. (2023). Pareto-improving seat allocation for high-speed railway networks with equilibrium flows. *Transportation Research Part C: Emerging Technologies*, 154, 104261.

[8] **Gao, Y.**, Yang, W., and Xu, G. (2022). An Optimization Method for Railway of Multi-train Seat allocation Problem with Cancellation and Refund Policy. *Journal of Railway Science and Engineering*, 19(3), 634-641. (In Chinese.)

BOOK CHAPTERS

[1] Liu, W., Wu, R., Xu, G., and **Gao, Y.** (2025). A double time-scale passenger assignment model. *Encyclopedia in Operations Management*, Vol. 2, pp. 224-237.

TEACHING EXPERIENCE

The Hong Kong Polytechnic University

AAE3009 Operations Research and Computational Analytics in Air Transport Operations

Teaching Assistant (tutorials) | Fall 2023, 2024, 2025 and 2026

AAE5102 Operations Research, Resource Planning and Engineering Management in Aviation

Teaching Assistant | 2024, Semester 1

AAE6106 Networked Transportation and Air Traffic Systems

Teaching Assistant | 2024, Semester 2

AAE3010 Airline Operations

Teaching Assistant | 2023, Semester 2

CONFERENCE PRESENTATIONS

[1] Robust planning of bus fleet electrification and charging facility deployment. *12th Triennial Symposium on Transportation Analysis (TRISTAN XII)*. Okinawa, Japan, Jun 22-27, 2025.

[2] Optimizing parking and charging services for electric autonomous vehicles considering vehicle-to-grid facilities. *World Transport Convention (WTC 2024)*. Qingdao, China, Jun 26-29, 2024.

[3] Joint optimization of vehicle-to-grid facilities and parking and charging services in the era of electric autonomous vehicles. *27th International Conference of Hong Kong Society for Transportation Studies (HKSTS 2023)*. Hong Kong, China, 2023.

[4] Integrated optimization of train stop planning and seat allocation for railway systems under equilibrium travel choice and elastic demand. *4th International Symposium on Multimodal Transportation*. Nanjing, China, 2021.

RESEARCH PROJECTS

[1] Integrated planning and management of electric vehicle travel, parking, charging and vehicle-to-grid services. General Program, Natural Science Foundation of Guangdong Province. 2023-2025. **Participant.**

[2] Joint optimization of seat allocation and differentiated ticket pricing for high-speed railway networks with time-varying demand. National Natural Science Foundation of China, No. 72171236. 2022.01-2025.12. **Participant.**

[3] Seat allocation optimization for high-speed railway networks with elastic time-varying demand. Postgraduate Research and Innovation Project, Central South University, No. 1053320192457. 2020.09-2022.03. **Project Lead.**

AWARDS AND HONORS

Commendation Award, Arup Research Competition 2025

Second Prize, 17th China Postgraduate Mathematical Contest in Modeling (Huawei Cup) 2020